

Rebuilding an LPG backend without stopping the business

Backend re-engineering, a pre-order dashboard and a mini app on the Safaricom M-Pesa app

85% to 98%

System uptime after the backend redesign

8%+

Increase in monthly recurring revenue from the pre-order dashboard and M-Pesa mini app

End to end

Ownership from prioritization through rollout

Role	Product Manager
Company	Proto Energy, which helps roadside retailers access Liquefied Petroleum Gas (LPG) cylinders on credit
Scope	End-to-end ownership of the redesign and the products built on it, across cross-functional teams
Skills used	Prioritization, stakeholder and vendor management, systems rollout, SOP documentation

The challenge

Proto Energy ran on a legacy backend that, in some months, managed only 85% uptime. That is 15% of operational hours lost, and with them significant revenue.

The problem ran deeper than downtime: anything built on the legacy infrastructure would be bottlenecked from scaling, so new features would only add weight to an unstable foundation.

Through ideation and prioritization, I concluded that a full backend redesign was the most business-critical work, ahead of any new feature. It was not a comfortable call: the redesign had many moving pieces, involved a factory floor handling dangerous items, and the business had to keep running while the system was rebuilt.

What I did

- Owned the redesign end to end and coordinated the cross-functional teams delivering it.
- Led the integration of third-party trackers on the LPG trucks as part of the redesign, and managed the relationship with the vendor, Fleetio.
- Managed a wide group of stakeholders with honest communication rather than lip service.
- Documented new SOPs and processes so teams could work confidently on the rebuilt systems.
- Once the foundation was stable, led the team to build a pre-order dashboard and a mini app launched on the Safaricom M-Pesa app.

The outcome

Uptime rose from 85% to 98%. With the foundation stable, the pre-order dashboard and the M-Pesa mini app raised monthly recurring revenue by more than 8%.

What I took from it

Fix the foundation before building on it. Reliability work rarely looks exciting on a roadmap, but it set the ceiling for every later feature. Honest stakeholder communication mattered as much as the plan.